

COLLAB • COLLABORATIVE DEVELOPMENT GROUP

Towards LLM-Driven State Machines for Runtime Robot Behavior in ROS 2

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Research question and hypothesis

RQ THIS WORK

Can an LLM govern the behaviour of a robot **reliably**?

WHAT IS ALREADY KNOWN

An LLM has **known problems**, and hallucination is one of them.

THE HYPOTHESIS

Raise reliability by **constraining the answer of the model with a schema**.

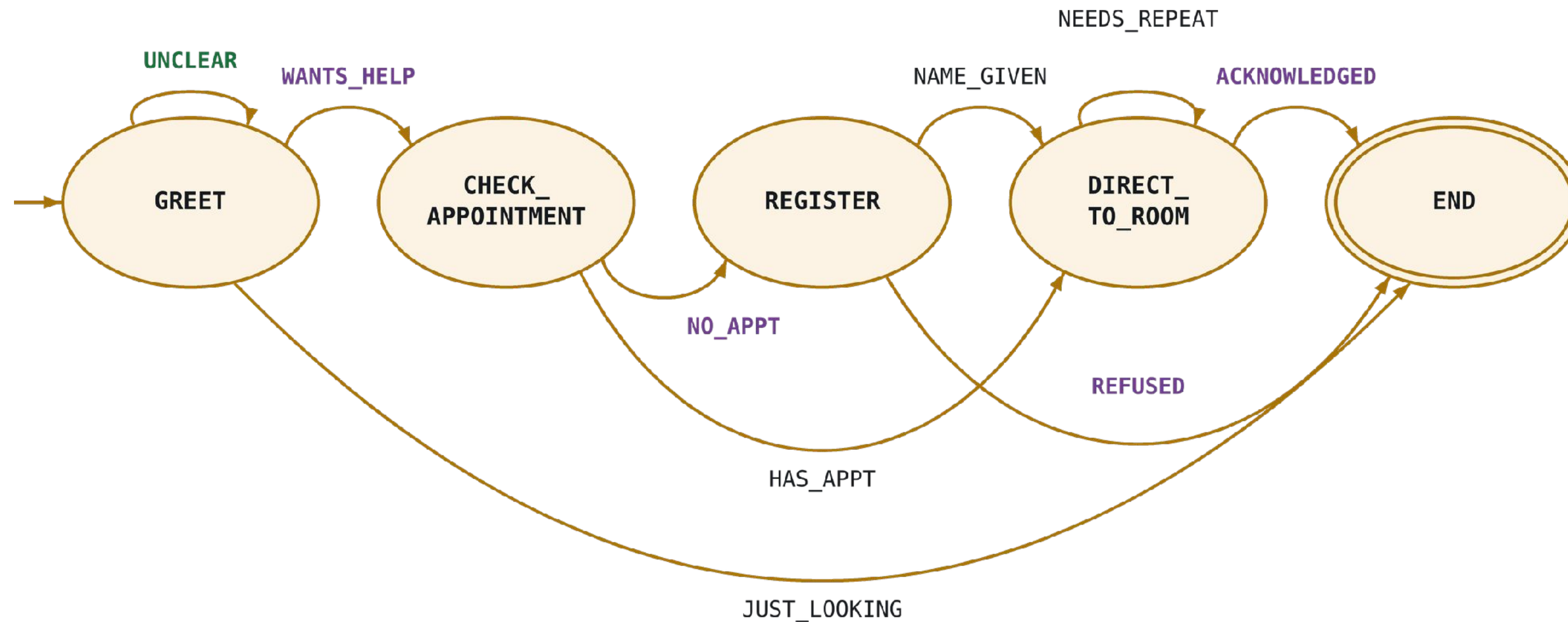
CONSTRAINTS

4B models only, so it runs on any Jetson, and offline, so the utterance never leaves the machine.



The reception, state by state

Held-out set: 20 cases, 35% ambiguous → **UNCLEAR self-loop** on every state, drawn once.

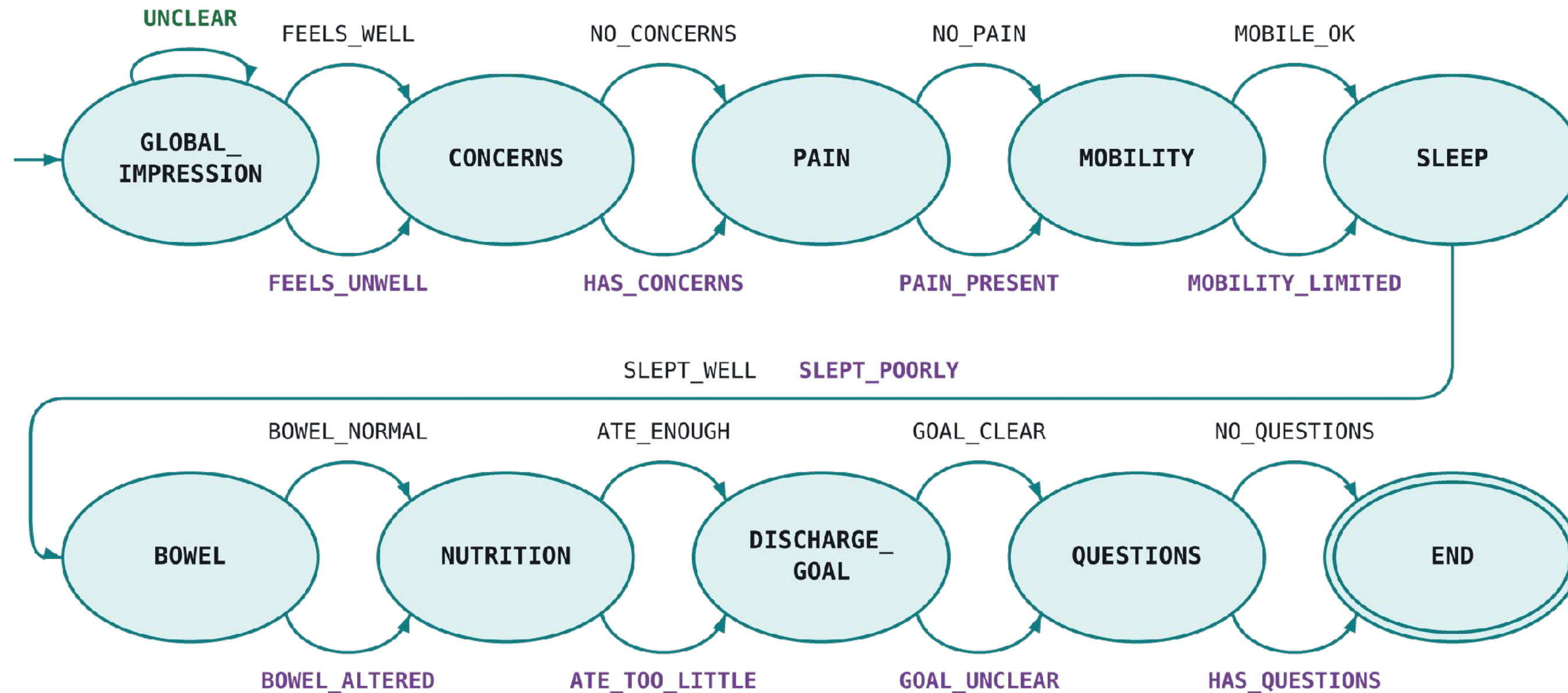


the safe_default



The ward round, state by state

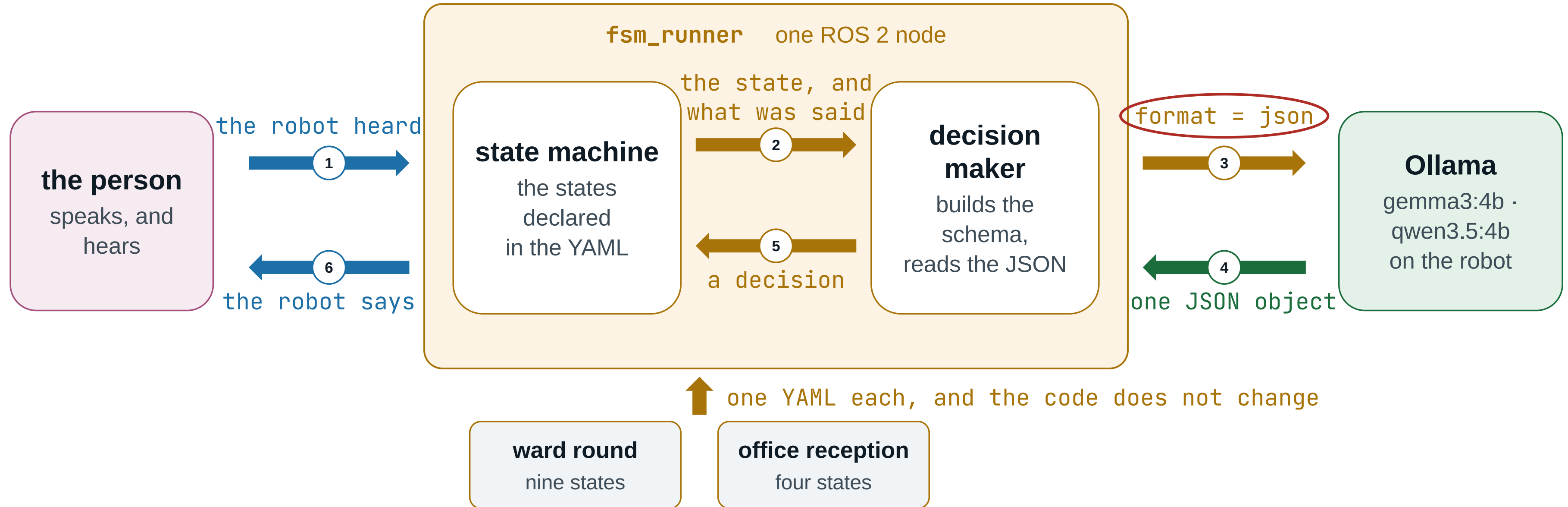
Held-out set: 60 cases, 32% ambiguous → **UNCLEAR self-loop** on every state, drawn once.



the safe_default



The system, end to end



the structure (what the system enforces) speech, in and out the external input the models



The unconstrained baseline

One call: understand the reply, say something back, choose the transition. **No schema at all.**

H0 the format field sent to Ollama

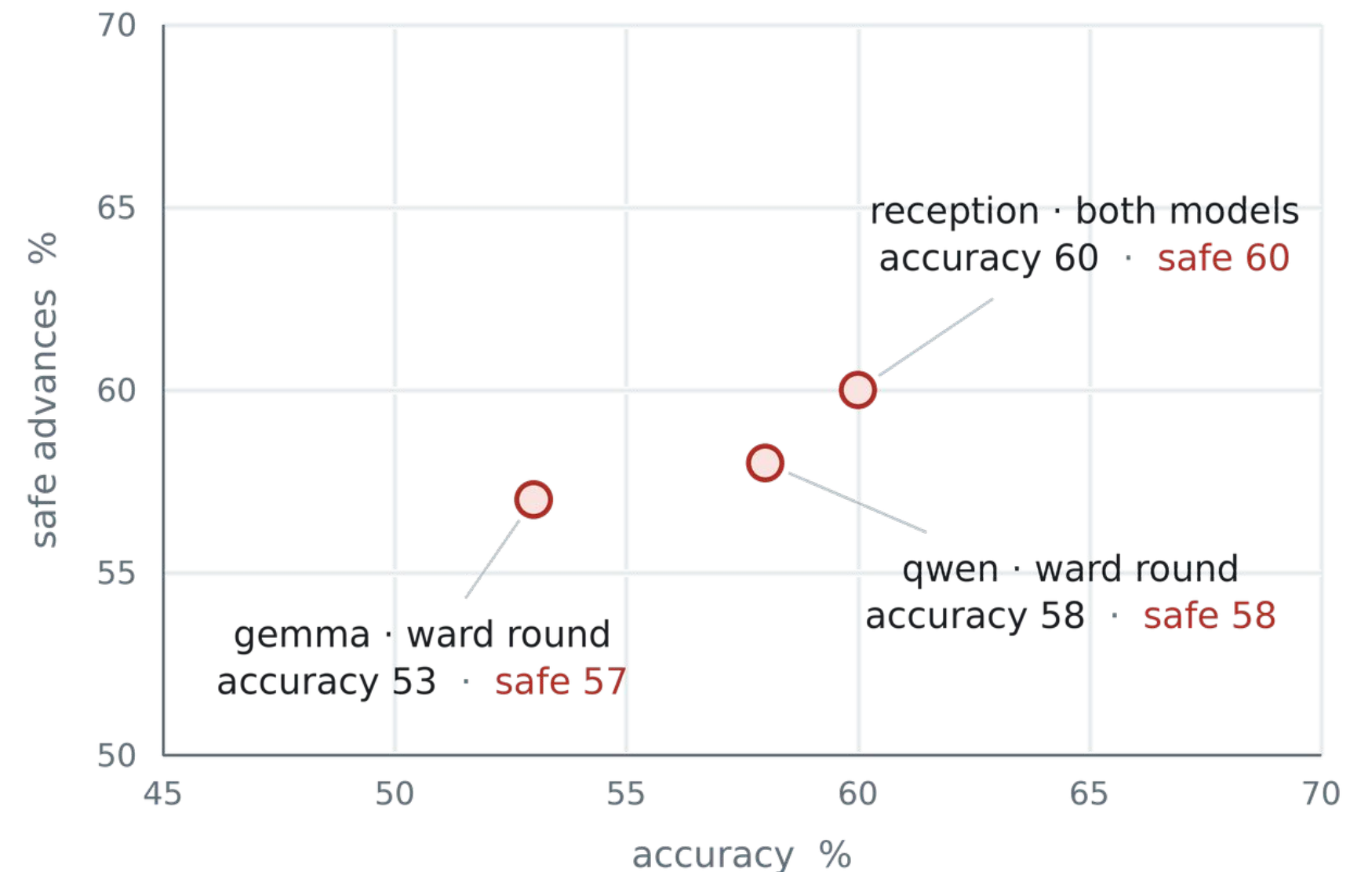
"json"

JSON mode: valid JSON, and nothing more.

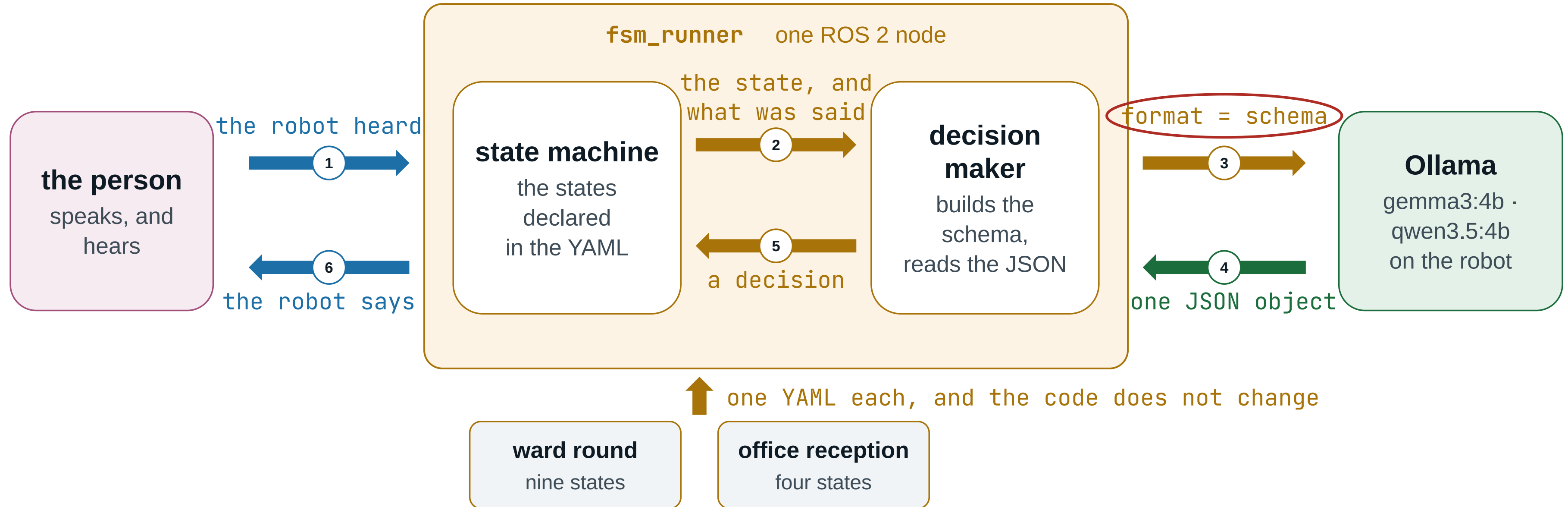
Which fields exist, and which outcomes exist, is written **only in the prompt**. So an outcome that does not exist is still representable, and the model produces one.

SAFE ADVANCES 57 · 58 · 60 · 60 %

and accuracy never above **60 %**



The system, end to end



Three schemas, six hypotheses

```
{ "response": string,  
  "outcome": enum [  
    ATE_ENOUGH,  
    ATE_TOO_LITTLE ] }
```

1

```
{ "evidence": string,  
  "outcome": enum [  
    ATE_ENOUGH,  
    ATE_TOO_LITTLE,  
    UNCLEAR ],  
  "confidence": number 0-1,  
  "followup": string }
```

2

```
{ "reasoning": string,  
  "evidence": string,  
  "outcome": enum [  
    ATE_ENOUGH,  
    ATE_TOO_LITTLE,  
    UNCLEAR ],  
  "confidence": number 0-1,  
  "followup": string }
```

3

H1	1	Is it choosing transitions that do not exist?	constrain the outcome to the enum
H2	2	Is one call being asked for too much at once?	classify only, and drop the spoken reply
H3	2	Does it need a way to say “I don’t know”?	stay in the state when the model says UNCLEAR
H4	3	Does it need to think before it chooses?	a reasoning field, before the outcome
H5	3	Does the thinking need guardrails?	instruct it to check the reply answers
H6	2 3	Do abstaining and classifying belong in one call?	two calls: abstain, then classify



Results of the hypotheses, H0 to H6

accuracy % / safe advances %

		ward · gemma	ward · qwen	recep · gemma	recep · qwen
H0		53 / 57	58 / 58	60 / 60	60 / 60
H1	vs H0	55 / 55	57 / 57	60 / 60	60 / 60
H2	vs H1	47 / 47	57-58 / 57-58	65 / 65	65 / 65
H3	vs H2	68 / 80	78 / 90	85 / 90	100 / 100
H4	vs H3	70-72 / 75-78	82-83 / 92	75 / 80	95 / 95
H5	vs H4	72-73 / 75-78	75 / 88-90	75 / 80	100 / 100
H6	vs H5	77 / 87	73 / 90	80-85 / 90	100 / 100

Abstention, together with constraining the answer of the model with a schema, **is what raised reliability.**



Field-by-field adherence: schema versus prompt

One call, one JSON object. Take H3: its four fields are governed from two different places.

`format = schema` in the Ollama request

`rules 2 and 4` in the system prompt

outcome	enum in the schema	✓ never outside the enum
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confidence	number, 0 to 1	✓ never invalid
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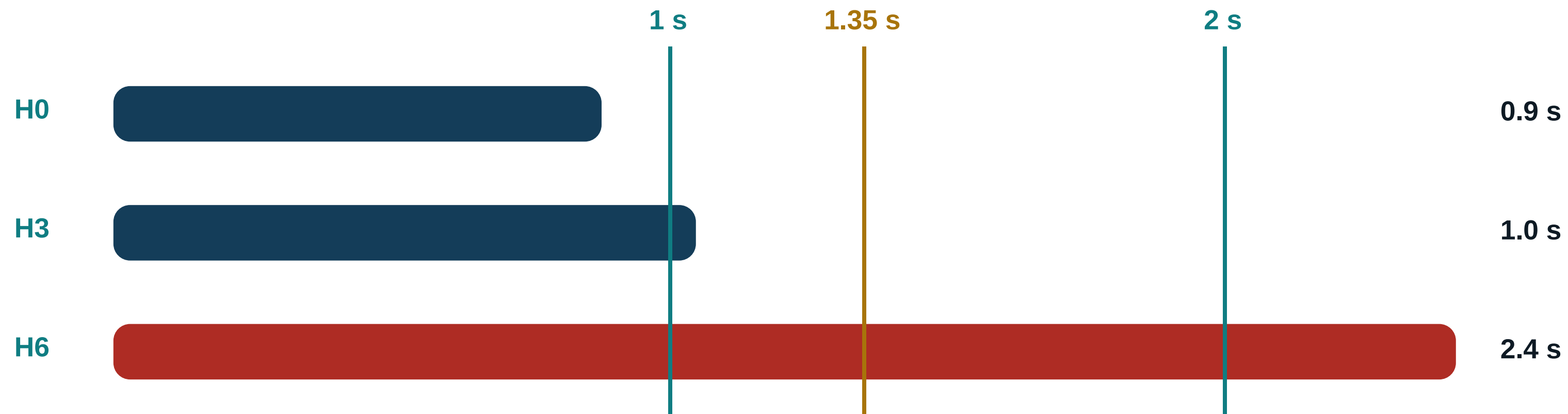
evidence	“copy VERBATIM”	✗ not a verbatim quote in 41% of calls
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followup	“use the user’s language”	✗ empty 62% · English · chases the off-topic
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What the schema enforces is obeyed. What the prompt asks for is not.



Latency against the human interaction budget



Median over both models and both domains. **And this is the model call alone, no ASR, no speech synthesis, no network.**

1 s · 2 s Shiwa et al., HRI 2008 **1.35 s** Abbo et al., Frontiers in Robotics and AI 2026



Conclusion and future work

CONCLUSION

- What the schema enforces is obeyed, what the prompt asks for is not.
- Abstention raises reliability, the schema alone does not.

FUTURE WORK

- Waiting on the physical robot
- Waiting on a second person that checks my transition labels



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